Review notes · for you

Data engineering: unit check

Data engineering · Level 4 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Schemas and formats
- ETL with Python and SQL
- Data quality and contracts
- Batch and stream

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (15 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (8 min): Reveal each answer and ask one team to explain why it's right.

Discussion

- Which part of a pipeline fails most often, and why?
- What would you put in a contract for data you depend on?
- Which system you use should be a stream, and isn't?

Check yourself

- One design decision I'd now make differently, and why.

Try it this week

Pick one dataset you know and design its whole path: the schema, the format, an idempotent load, three contract tests and batch or stream.

Quiz sheet

Data engineering: unit check

Data engineering · Level 4 · Unit check

Name _____ Class _____ Date _____

1 **In a star schema, where does each school's name belong?**

◯ In the run log ◯ In every row of the fact table ◯ Once, in the schools dimension table

2 **JSON is usually more than twice the size of CSV for the same table, because it repeats every column's name.**

◯ True ◯ False

3 **Put the steps of a safe nightly load in order.**

☐ Delete the day's rows, then insert

☐ Transform them

☐ Read the watermark

☐ Save the new watermark

☐ Extract the newer rows

4 **Fill the gaps: an idempotent load of today's rows.**

```
_________ FROM daily WHERE day = 'Mon';
_________ INTO daily SELECT * FROM today;
```

Word bank: `SELECT` `UPDATE` `DELETE` `INSERT`

5 **Match each quality test to what it catches.**

Range	• • • •	Today's file arriving at 09:30, not 07:00
Referential integrity	• • • •	A mark of 25 out of 20
Freshness	• • • •	The same reading loaded twice
Uniqueness	• • • •	A mark for pupil 7 when there's no pupil 7

6 **This orphan test never returns a row, even when there are orphans. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT marks.pupil_id
2  FROM marks
3  JOIN pupils
4    ON pupils.id = marks.pupil_id
5  WHERE pupils.id IS NULL;
```

---

7 **A test fails on 3 rows out of 10,000 in a daily batch. What's usually the best response?**

◯ Block the whole batch ◯ Delete the 3 rows quietly

◯ Quarantine the 3 rows and alert the producer

8 **Which of these need a stream rather than a nightly batch? Choose all that apply.**

☐ The year's exam results ☐ The monthly attendance report ☐ A card fraud alert

☐ A flood warning

9 **Complete the sentence.**

A bigger averaging window gives _____ false alarms, but the real alert comes _____ .

Word bank: more · later · fewer · sooner

10 **What does this program print?**

```
print("9" > "20")
```

◯ False ◯ True ◯ An error

Check your answers

Data engineering: unit check

Data engineering · Level 4 · Unit check

1 **Once, in the schools dimension table**

Dimensions hold descriptions once; the fact table holds only keys and measures, so it stays small however many events it has.

2 **True**

Every JSON record spells out the column names again; CSV says them once in the header.

3 **Top to bottom: 4, 3, 1, 5, 2**

The watermark decides what's extracted and is saved last, after a successful load, so a failure simply reruns; delete-then-insert makes that rerun safe.

4

```
DELETE FROM daily WHERE day = 'Mon';
INSERT INTO daily SELECT * FROM today;
```

Delete the day's rows, then insert them again: one run or five leaves the same table.

5 **Range — A mark of 25 out of 20**
Referential integrity — A mark for pupil 7 when there's no pupil 7
Freshness — Today's file arriving at 09:30, not 07:00
Uniqueness — The same reading loaded twice

Each test checks one side of quality; together they cover most of what a contract promises.

6 **Line 3: `LEFT JOIN pupils`**

An inner join drops the orphans before WHERE sees them, so the test always passes.

7 **Quarantine the 3 rows and alert the producer**

Good data keeps flowing, and the bad rows wait, flagged, for a fix at the source.

8 **A flood warning; A card fraud alert**

Floods and fraud can't wait; reports and results can, and a batch gives them more care for less cost.

9 **A bigger averaging window gives fewer false alarms, but the real alert comes later.**

More readings smooth out the noise, so splashes stop alarming, but a real rise takes longer to lift the average.

10 **True**

Text is compared letter by letter, and "9" comes after "2", so "9" > "20" is True.